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$9z^2 + 3(2i-5)z + (9-7i) = 0$

$a = 9, \quad b = 6i - 15, \quad c = 9 - 7i$

$b^2 = (6i-15)^2 = 36i^2 - 180i + 225 = -36 - 180i + 225 = 189 - 180i$

$4ac = 4 \cdot 9 \cdot (9-7i) = 36(9-7i) = 324 - 252i$

$\Delta = b^2 - 4ac = (189 - 180i) - (324 - 252i) = -135 + 72i$

$(x+yi)^2 = -135 + 72i \quad \Rightarrow \quad x^2 - y^2 = -135, \quad 2xy = 72$

$xy = 36, \quad y = \dfrac{36}{x}$

$x^2 - \left(\dfrac{36}{x}\right)^2 = -135 \quad \Rightarrow \quad x^4 + 135x^2 - 1296 = 0$

$u = x^2$: $u^2 + 135u - 1296 = 0$, $u = 9$ of -144

$x^2 = 9 \quad \Rightarrow \quad x = 3, \quad y = \dfrac{36}{3} = 12$

$\sqrt{\Delta} = \pm(3 + 12i)$

$z = \dfrac{-b \pm \sqrt{\Delta}}{2a} = \dfrac{15 - 6i \pm (3+12i)}{18}$

$z_1 = \dfrac{18 + 6i}{18} = 1 + \dfrac{1}{3}i$

$z_2 = \dfrac{12 - 18i}{18} = \dfrac{2}{3} - i$

$$\boxed{z_1 = 1 + \frac{1}{3}i \quad , \quad z_2 = \frac{2}{3} - i}$$

References